Table of Contents

1. Deviation — Definition and Sign Convention
2. Compass Swing — Purpose and Procedure
3. Hard Iron Magnetism
4. Soft Iron Magnetism (VSI)
5. Coefficients A, B, and C
6. Correction of Coefficients
7. Accuracy Limits
8. Occasions for Swinging the Compass
9. Practice Questions & Detailed Answers
10. Master Reference Tables

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1. Deviation — Definition and Sign Convention

> **What this section covers:** Definition of deviation; east/west naming; mnemonic for compass vs. magnetic heading.

Deviation is the angular difference measured between the direction taken up by a compass needle and the magnetic meridian, caused by the aircraft's own internal magnetism deflecting the needle.

Deviation is named **easterly** or **westerly** depending on whether the north-seeking end of the compass needle lies to the east or west of the magnetic meridian.

Deviation	Compass Heading (C)	Sign	Magnetic Heading (M)	Mnemonic
West	095	−5	090	**Deviation West, Compass Best**
East	090	+5	095	**Deviation East, Compass Least**

M = C + Deviation (east positive, west negative)

Or equivalently: C = M − Deviation

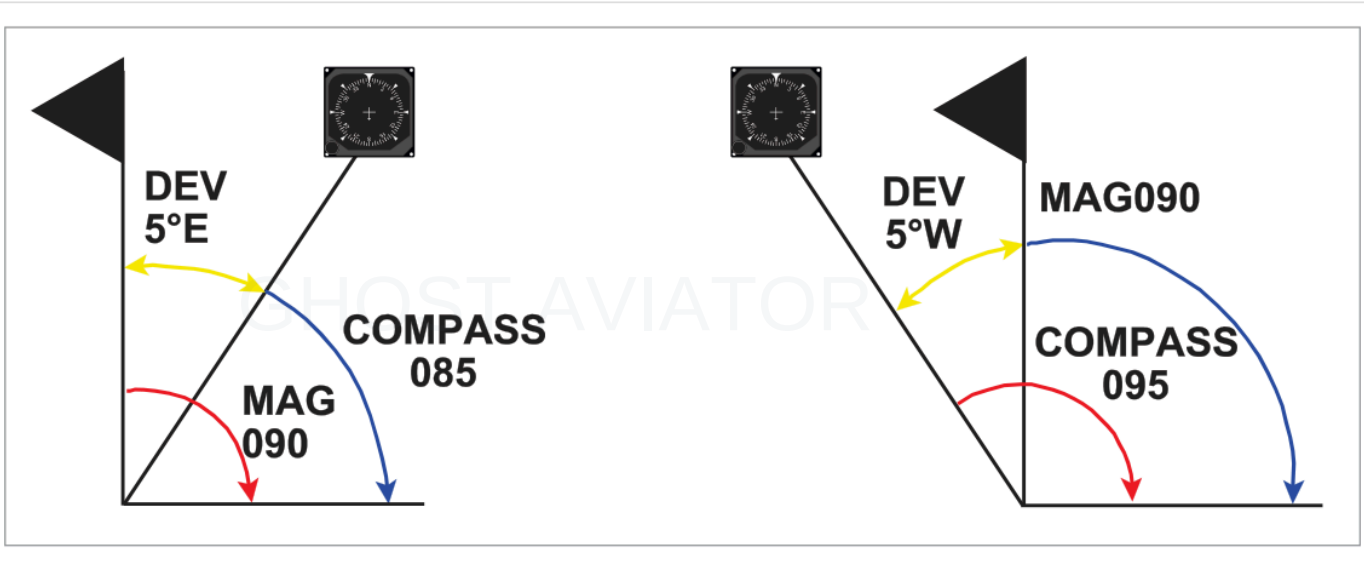


Fig 16.1 — Deviation illustration: compass heading vs. magnetic heading.

2. Compass Swing — Purpose and Procedure

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What this section covers: What a compass swing achieves; the three aims; where it is conducted.

A **compass swing** is the basic method of determining deviation by comparing the aircraft's heading compass reading with magnetic heading as defined by a high quality 'land or datum' compass, carried out in an area specifically selected for this purpose.

Three Aims of a Compass Swing:

1. To **observe/determine** the deviations/differences between magnetic north (landing compass) and compass north (aircraft compass) on a series of headings.
2. To **correct/remove** as much deviation as possible.
3. To **record the residual deviation** remaining after adjustment — producing a Compass Deviation Card placed near the compass.

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The magnetic deviation observed during a compass swing is derived from **hard iron** and **soft iron** magnetism, resolved into two combined components (coefficients B and C).

3. Hard Iron Magnetism

What this section covers: Nature and properties of hard iron magnetism.

Hard iron magnetism consists of **permanent magnets** in the aircraft structure — parts that retain their magnetism independently of any external field. The total force at the compass position produced by hard iron magnetism can be resolved into three components (longitudinal, lateral, and vertical). These components are **fixed for a given aircraft** and do *not* change with change of heading.

Key Property of Hard Iron: The magnetic components due to hard iron are constant regardless of the aircraft's heading or latitude.

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4. Soft Iron Magnetism (Vertical Soft Iron)

What this section covers: What soft iron magnetism is; how it varies with latitude; the role of the earth's vertical component Z.

Soft iron magnetism is **induced** in parts of the aircraft structure by surrounding fields — most importantly the earth's magnetic field. Unlike hard iron, soft iron magnetism changes with the surrounding field.

The earth's field has both a vertical component (Z) and a horizontal component (H). Within the examination syllabus, we consider primarily **Vertical Soft Iron (VSI)** magnetism, induced by component Z:

- Z increases with latitude (as the earth's field dips more steeply) → VSI magnetism **increases with latitude**.
- Z is zero at the equator → **no VSI magnetism is induced at the equator**.
- VSI magnetism varies with magnetic latitude — but unlike hard iron, it is *not* fixed with heading.

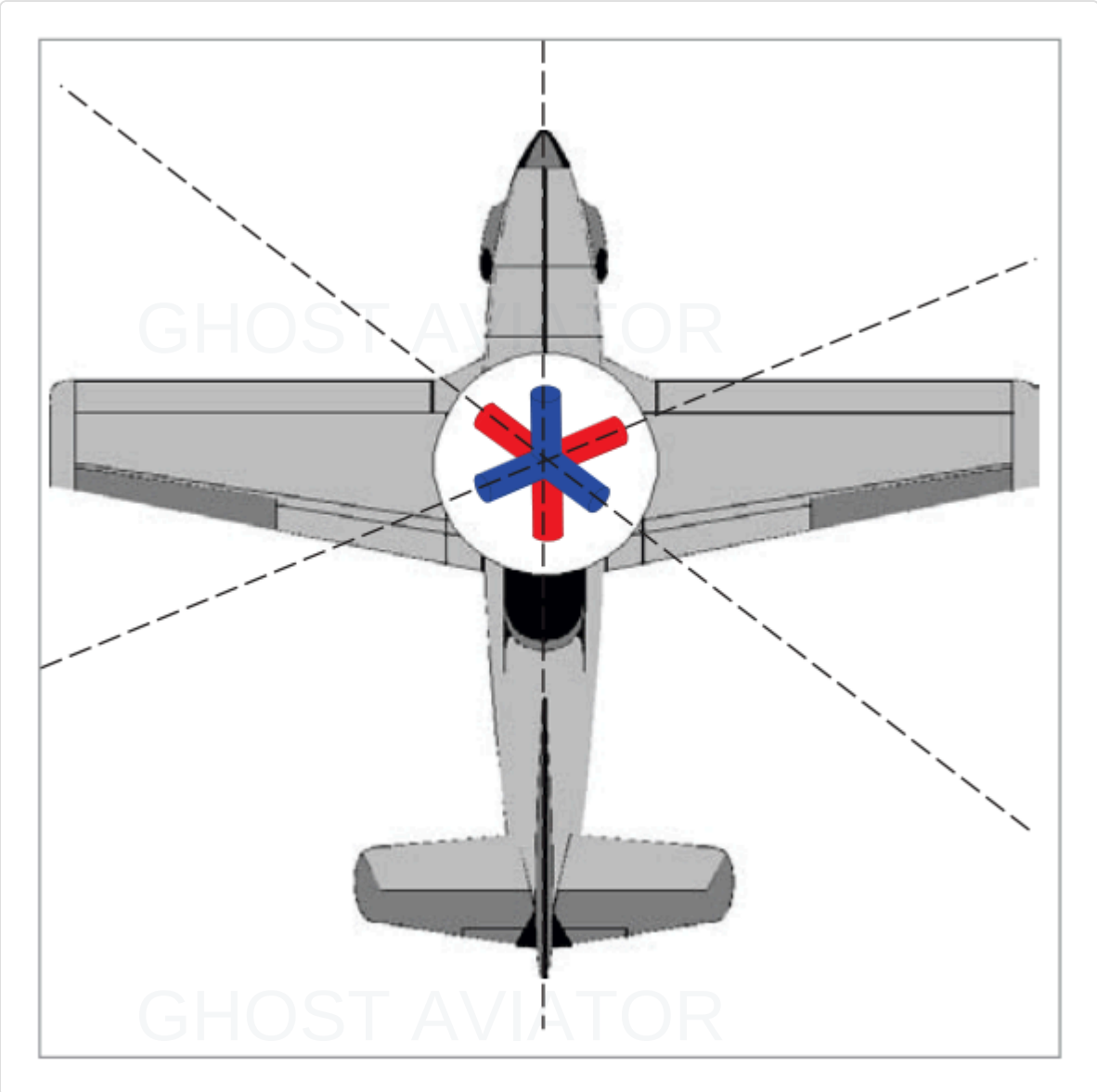


Fig 16.2 — Hard iron and soft iron magnetic forces on the compass.

5. Coefficients A, B, and C

> **What this section covers:** What each coefficient represents and the pattern of deviation it produces.

The deviations observed during a compass swing are resolved into three coefficients:

Coefficient	Cause	Deviation Pattern	Effect
A	Mechanical misalignment of lubber line (or compass body)	Constant on all headings	Same deviation on all headings
B	Magnetic deviating forces along the fore-aft axis (e.g. blue pole in nose or tail)	**Sine curve**: max on E/W, zero on N/S	Blue pole in nose → max +E deviation on 090°, max W on 270°
C	Magnetic deviating forces along the lateral axis (e.g. pole to port/starboard)	**Cosine curve**: max on N/S, zero on E/W	Force resolved to right wing → positive cosine curve

Exam Tip — Sine vs. Cosine:

- Coefficient B (fore-aft pole) → **sine curve** → zero deviation on N and S; max on E and W.
- Coefficient C (lateral pole) → **cosine curve** → zero deviation on E and W; max on N and S.

6. Correction of Coefficients

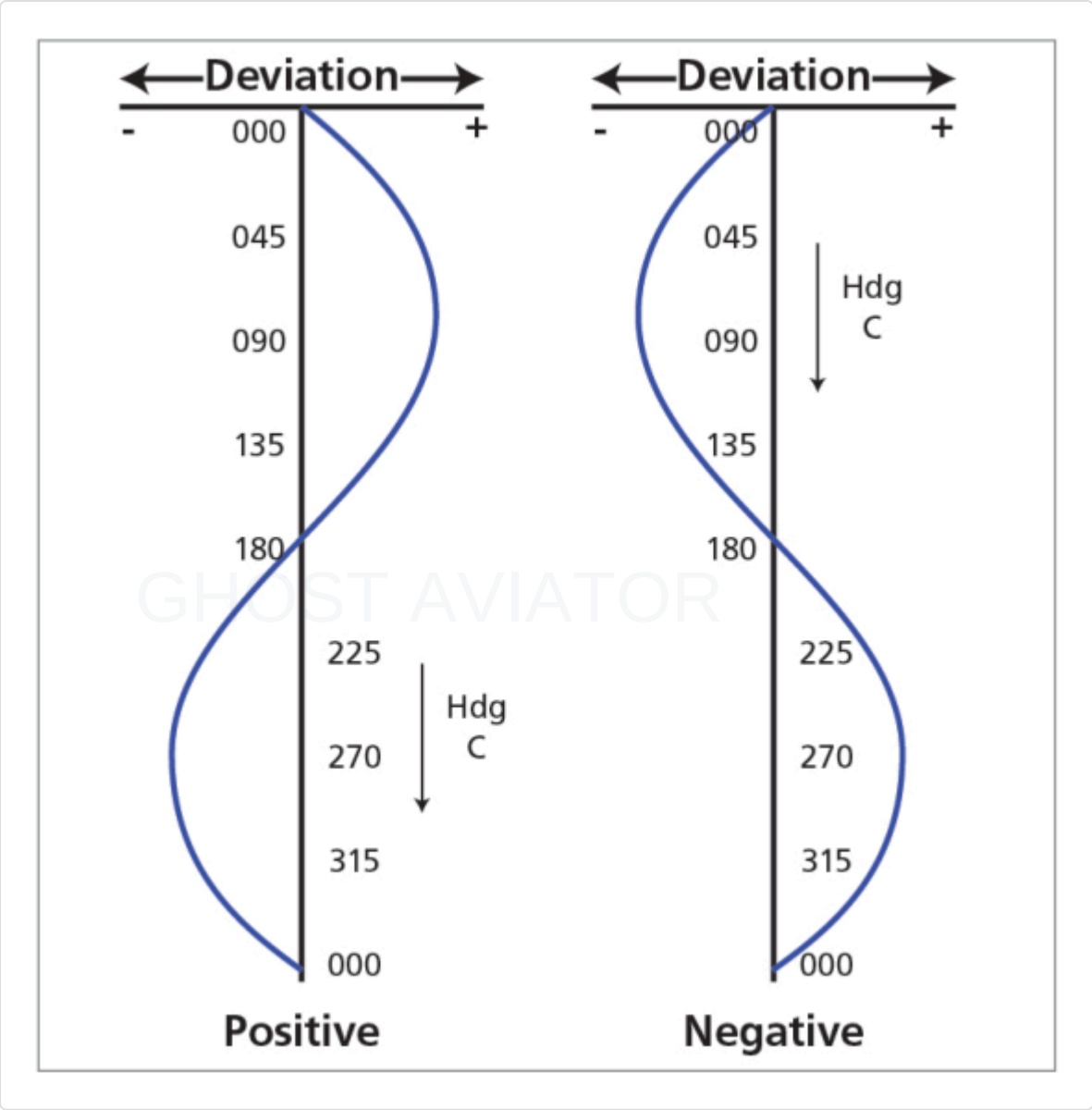
What this section covers: How each coefficient is corrected during a compass swing.

Correction Method for Each Coefficient:

- **Coefficient A**: A mechanical problem (displaced lubber line). Corrected by loosening the bolts holding the compass body (or detector unit in RIMC) and carefully turning it to align correctly.
- **Coefficient B**: Corrected by adjusting compensating magnets on an easterly or westerly heading (where B deviation is maximum). Calculate the heading you wish the compass to read and adjust accordingly.
- **Coefficient C**: Corrected by adjusting compensating magnets on a northerly or southerly heading (where C deviation is maximum). Similar calculation and sign application as for B.

After correction for B and C, a **check swing** is carried out using eight or twelve points of the compass to verify the work and derive the residual deviation for the **Compass Deviation Card**.

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Residual deviations after compass swing — represented as a table, curve, or compass deviation card.

7. Accuracy Limits

What this section covers: The regulatory maximum permissible deviations after correction.

REGULATORY LIMITS (CS OPS-1 / EU OPS-1):

Compass Type	Maximum Residual Deviation After Correction
Direct Reading Magnetic Compass (DRMC)	**±10°**
Remote Indicating Compass (RIMC)	**±1°**

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8. Occasions for Swinging the Compass

What this section covers: All regulatory occasions requiring a compass swing.

A compass swing is required when:

- Compass components are installed or replaced.
- Whenever the accuracy of the compass is in doubt.
- After a maintenance inspection if required by the schedule.
- After a significant aircraft modification, repair or replacement involving magnetic material.
- When carrying unusual ferromagnetic payloads.
- When the compass has been subjected to significant shock.
- If the aircraft has been struck by lightning.
- After significant modification to aircraft radio/electrical systems.
- After the aircraft has been given a new theatre of operations if the move involves a large change of magnetic latitude.
- If the aircraft has been in long-term storage standing on one heading.

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flowchart TD
    SW[Compass Swing Required?] --> A[New/replaced\ncompass components?]
    SW --> B[Accuracy\nin doubt?]
    SW --> C[Maintenance\ninspection\nschedule?]
    SW --> D[Magnetic material\nmodification/repair?]
    SW --> E[Ferromagnetic\npayload?]
    SW --> F[Significant\nshock or\nlightning strike?]
    SW --> G[Radio/electrical\nsystem mods?]
    SW --> H[New theatre:\nlarge latitude\nchange?]
    SW --> I[Long-term storage\non one heading?]
    A --> YES[YES → SWING]
    B --> YES
    C --> YES
    D --> YES
    E --> YES
    F --> YES
    G --> YES
    H --> YES
    I --> YES
```

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Quick Revision Summary — Chapter 16:

- Deviation = angle between compass needle and magnetic meridian. East = positive (+); West = negative (−).
- Mnemonic: "Deviation West, Compass Best" (C > M). "Deviation East, Compass Least" (C < M).
- Compass swing done at a dedicated compass swinging area using a landing/datum compass.
- Three aims: Observe → Correct → Record residual (deviation card).
- Hard iron: permanent magnetism, fixed with aircraft, not affected by heading or latitude.
- Soft iron (VSI): induced by earth's field, varies with latitude (max at poles, zero at equator).
- Coefficient A: constant deviation, mechanical misalignment.
- Coefficient B: sine curve deviation, corrected on E/W headings.
- Coefficient C: cosine curve deviation, corrected on N/S headings.
- DRMC accuracy limit: **±10°**. RIMC accuracy limit: **±1°**.

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Practice Questions & Detailed Answers

Source questions reproduced verbatim. Answer key from source: 1-a, 2-b, 3-b, 4-a, 5-c, 6-d, 7-b.

Q1. European regulations (EU OPS-1) state that the maximum permissible deviations after compensation are:

a. one degree for a remote indicating compass, and ten degrees for a direct reading magnetic compass

b. three degrees for a direct reading magnetic compass, and one degree for a remote indicating compass

c. ten degrees for a remote indicating compass, and one degree for a direct reading magnetic compass

d. one degree for a direct reading magnetic compass, and eleven degrees for a slaved compass

Correct Answer: **(a) 1° for remote indicating compass; 10° for direct reading magnetic compass**

Explanation: CS OPS-1 / EU OPS-1 specifies: DRMC = ±10° maximum residual deviation; RIMC = ±1° maximum residual deviation. See Section 7.

Why the other options are wrong:

- **(b)** — Reverses the limits (3° for DRMC is wrong; it is 10°). And 1° for RIMC is correct but the DRMC value is wrong.
- **(c)** — Reverses the instrument types. 10° applies to DRMC, not RIMC.
- **(d)** — Neither figure is correct for a direct reading compass (should be 10°, not 1°), and 11° is not a regulatory value.

Instructor's Note: *RIMC = 1° (very tight, because it is a precision instrument feeding autopilots and FMS). DRMC = 10° (larger tolerance for a simpler instrument). Easy mnemonic: "Remote = 1; Direct = 10."*

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Q2. Compass swings should be carried out:

a. on the apron

b. only on the compass swinging base or site

c. at the holding point

d. on the active runway

Correct Answer: **(b) only on the compass swinging base or site**

Explanation: The compass swing must be conducted in an area specifically selected for this purpose — the compass swinging base or site — which is chosen to be free from magnetic interference (underground cables, reinforcing, aircraft equipment). See Section 2.

Why the other options are wrong:

- **(a)** — The apron has underground infrastructure (fuelling pipes, cables) that cause magnetic interference.
- **(c)** — The holding point is on or near the runway, which has metallic reinforcing in the concrete/tarmac.
- **(d)** — The runway itself is metallic reinforced concrete, highly unsuitable for compass swinging.

Instructor's Note: *The compass swinging base is a specially prepared area on the airfield, away from hangars, pipelines, and other magnetic interference. The landing/datum compass used must also be calibrated and away from interference.*

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Q3. Aircraft magnetism caused by vertical soft iron:

a. varies with magnetic heading but not with magnetic latitude

b. varies with magnetic latitude but not with heading

c. is not affected by magnetic latitude or heading

d. varies as the cosine of the compass heading

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Correct Answer: **(b) varies with magnetic latitude but not with heading**

Explanation: Vertical Soft Iron (VSI) magnetism is induced by the vertical component Z of the earth's field. Z increases with latitude (zero at equator, maximum at poles). So VSI magnetism varies with magnetic latitude. However, for a given latitude, the component Z is the same regardless of which way the aircraft is pointing — so VSI magnetism does not vary with heading. See Section 4.

Why the other options are wrong:

- **(a)** — VSI does vary with latitude (not just heading). This is the opposite of correct.
- **(c)** — VSI is absolutely affected by latitude (Z varies with latitude).
- **(d)** — VSI does not vary as cosine of heading; it is a function of latitude, not heading.

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Instructor's Note: *Compare with hard iron (constant regardless of heading OR latitude) and soft iron (varies with latitude). This distinction is a frequent exam topic.*

Q4. Aircraft magnetism caused by hard iron:

a. is not usually influenced by the earth's magnetic field

b. varies directly with magnetic latitude

c. varies indirectly with magnetic latitude

d. is maximum on east and west

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Correct Answer: **(a) is not usually influenced by the earth's magnetic field**

Explanation: Hard iron magnetism is due to *permanent* magnets — parts of the aircraft that retain their magnetism independently. They are not induced by the earth's field and therefore are not influenced by changes in heading or latitude. Their magnetic force is constant and fixed with the aircraft. See Section 3.

Why the other options are wrong:

- **(b) and (c)** — Varying with latitude (directly or indirectly) is a property of soft iron magnetism (specifically VSI), not hard iron.

- **(d)** — "Maximum on east and west" describes Coefficient B deviation pattern (sine curve), not the nature of hard iron itself.

Instructor's Note: *Hard iron = permanent, constant, not affected by external fields. Soft iron = induced, varies with the inducing field (earth's field). Clear contrast.*

Q5. The aim of a compass swing is:

1. to find deviation on the cardinal headings and to calculate coefficients A, B and C

2. to eliminate or reduce the coefficients found

3. to record any residual deviation and to prepare a compass correction card

a. only answer 1 is correct

b. answers 1 and 3 are correct

c. answers 1, 2 and 3 are all correct

d. none of the above answers are correct

Correct Answer: **(c) answers 1, 2, and 3 are all correct**

Explanation: All three statements describe the complete three-step aim of a compass swing: observe and calculate (1), correct (2), and record residual (3). All three are correct. See Section 2.

Why the other options are wrong:

- **(a)** — Incomplete; stops at observation only.
- **(b)** — Incomplete; omits the correction step (2), which is the central purpose.
- **(d)** — All three statements are correct.

Instructor's Note: *Three steps: Observe → Correct → Record. All three are mandatory parts of a complete compass swing.*

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Q6. Deviation due to coefficient A is mainly caused by:

a. hard iron force acting along the longitudinal axis

b. hard and soft iron forces acting along the lateral axis

c. vertical soft iron forces

d. a misaligned lubber line

Correct Answer: **(d) a misaligned lubber line**

Explanation: Coefficient A is a constant deviation on all headings, mainly caused by a mechanical misalignment of the lubber line (or the compass body/detector unit is not correctly aligned with the fore-aft axis of the aircraft). See Section 5 and Section 6.

Why the other options are wrong:

- **(a)** — Hard iron along the longitudinal axis causes Coefficient B (sine curve deviation).
- **(b)** — Lateral forces cause Coefficient C (cosine curve deviation).
- **(c)** — VSI causes deviation that varies with latitude — not a constant (Coefficient A) effect.

Instructor's Note: *Coefficient A = mechanical, not magnetic. B = fore-aft magnetic (sine). C = lateral magnetic (cosine). A is corrected physically (rotate compass body), B and C magnetically (compensating magnets).*

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Q7. In the diagram below, the compass heading of the aircraft is ....., the magnetic heading ..... and the true heading .....

a. 025° — 015° — 020°

b. 335° — 035° — 020°

c. 335° — 340° — 035°

d. 025° — 015° — 340°

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Correct Answer: **(b) Compass 335°, Magnetic 035°, True 020°**

Explanation: From the diagram (not reproduced here but standard format), three north indicators show compass north, magnetic north, and true north. The readings of each reference line against the aircraft heading indicator give the three heading values. The source confirms: C = 335°, M = 035°, T = 020°. See Section 1 for conversion between heading types.

Why the other options are wrong:

- **(a)** — The C, M, T values are all reversed or incorrect based on the diagram geometry.
- **(c)** — The compass and magnetic values are swapped.
- **(d)** — The compass value (025°) and the true value (340°) do not match the expected deviation sequence in the diagram.

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Instructor's Note: *In heading diagrams, read each north reference against the aircraft's heading line. The compass north shows compass heading; magnetic north shows magnetic heading; true north shows true heading.*

Master Reference Tables

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Parameter	Value / Property	Section
DRMC max residual deviation	**±10°**	7
RIMC max residual deviation	**±1°**	7
Hard iron variation with heading	Constant (no change)	3
Hard iron variation with latitude	Constant (no change)	3
VSI variation with latitude	Increases (zero at equator, max at poles)	4
VSI variation with heading	Does not vary with heading	4
Coefficient A: cause	Misaligned lubber line / compass body	5
Coefficient B: pattern	Sine curve; max deviation on E/W headings	5
Coefficient C: pattern	Cosine curve; max deviation on N/S headings	5
Coefficient B correction heading	E or W (deviation is maximum)	6
Coefficient C correction heading	N or S (deviation is maximum)	6
Deviation formula (E pos, W neg)	M = C + Dev	1

Answer Key Summary

Q	Answer	Key Topic
1	a	RIMC = ±1°; DRMC = ±10°
2	b	Compass swing only at compass swinging base/site
3	b	VSI varies with latitude; not with heading
4	a	Hard iron: not influenced by earth's field
5	c	All three aims of compass swing are correct
6	d	Coefficient A = misaligned lubber line
7	b	C=335°, M=035°, T=020°

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